

NEE24_Analysis

2025-09-19

```
library(tidyr)
library(nloptr)
library(nlme)
library(parallel)
```

Data

Loading Phylogenetic and Spatial Distance Matrices

```
phylomatrix <- read.csv("data/distance_matrices/phylogenetic_covariance_matrix.csv",
                        check.names = FALSE)
rownames(phylomatrix) <- phylomatrix[,1]
phylomatrix <- phylomatrix[,-1]
phylomatrix <- as.matrix(phylomatrix)
spmatrix <- read.csv("data/distance_matrices/spatial_distance_matrix.csv",
                     check.names = FALSE)
rownames(spmatrix) <- spmatrix[,1]
spmatrix <- spmatrix[,-1]
spmatrix <- as.matrix(spmatrix)
```

Preview Matrices

```
print(phylomatrix[1:5,1:5])
```

```
##          aaa      aab aac aad aae
## aaa 1.0000 0.0782  0  0  0
## aab 0.0782 1.0000  0  0  0
## aac 0.0000 0.0000  1  0  0
## aad 0.0000 0.0000  0  1  0
## aae 0.0000 0.0000  0  0  1
```

```
print(spmatrix[1:5,1:5])
```

```
##          aaa      aab      aac      aad      aae
## aaa      0.0000  348.5906 15210.7391 15144.2299 3664.13
## aab      348.5906   0.0000 14934.7335 14849.6166 3391.56
## aac 15210.7391 14934.7335   0.0000  385.0299 13574.52
## aad 15144.2299 14849.6166  385.0299   0.0000 13282.30
## aae  3664.1302  3391.5598 13574.5178 13282.3037   0.00
```

Loading NEE24 (1)

```
nee24 <- read.csv("data/Islands are engines of language diversity data.csv")
```

NEE24 Cleanup

```
# --- Remove all NA entries ---
paste("Size of initial dataset:", dim(nee24)[1])

## [1] "Size of initial dataset: 7175"
paste("#Island_Endemic: ", sum(nee24$Island.Endemic))

## [1] "#Island_Endemic: 630"
paste("#Island_Endemic to Total Ratio: ", sum(nee24$Island.Endemic) / dim(nee24)[1])

## [1] "#Island_Endemic to Total Ratio: 0.0878048780487805"
old_nee24 <- nee24
nee24 <- na.omit(nee24)
paste("Number of incomplete (NA) language entries removed:",
      dim(old_nee24)[1] - dim(nee24)[1])

## [1] "Number of incomplete (NA) language entries removed: 5410"
paste("Size of dataset with NA remove:", dim(nee24)[1])

## [1] "Size of dataset with NA remove: 1765"
paste("#Island_Endemic: ", sum(nee24$Island.Endemic))

## [1] "#Island_Endemic: 157"
paste("#Island_Endemic to Total Ratio: ", sum(nee24$Island.Endemic) / dim(nee24)[1])

## [1] "#Island_Endemic to Total Ratio: 0.0889518413597734"
# --- Convert variable types ---
nee24$Island.Endemic <- as.numeric(nee24$Island.Endemic)

# --- Log transform variables ---
nee24$Phoneme.Inventory.Size <- log(nee24$Phoneme.Inventory.Size)
nee24$Range.Size..km2. <- log(nee24$Range.Size..km2.)
nee24$L1.Population <- log(nee24$L1.Population + 0.5)
nee24$Distance.to.Mainland <- log(nee24$Distance.to.Mainland + 0.5)
nee24$Distance.to.Continent <- log(nee24$Distance.to.Continent + 0.5)
```

Preview data

```
head(nee24)

##      ISO693.3 Range.Size..km2. L1.Population Island.Endemic
## 9          aak          5.524229      7.378071            0
## 15         aaq          10.537220      7.495820            0
## 16         aar          11.809747     14.137015            0
## 19         aau           7.386708      8.891580            0
## 34         abk           8.409938     11.561720            0
## 36         abm           5.118227      9.433524            0
##      Phoneme.Inventory.Size Distance.to.Mainland Distance.to.Continent
## 9              2.944439          -0.6931472          -0.6931472
## 15              2.944439          -0.6931472          -0.6931472
## 16              3.218876          -0.6931472          -0.6931472
```

```
## 19          2.833213          -0.6931472          -0.6931472
## 34          3.583519          -0.6931472          -0.6931472
## 36          3.218876          -0.6931472          -0.6931472
```

```
summary(nee24)
```

```
##      ISO693.3      Range.Size..km2. L1.Population      Island.Endemic
## Length:1765      Min.      :-3.391      Min.      :-0.6931      Min.      :0.00000
## Class :character  1st Qu.: 6.226      1st Qu.: 7.1472      1st Qu.:0.00000
## Mode  :character  Median : 7.702      Median : 9.3927      Median :0.00000
##                               Mean  : 7.774      Mean  : 9.2779      Mean  :0.08895
##                               3rd Qu.: 9.396      3rd Qu.:11.8636      3rd Qu.:0.00000
##                               Max.   :14.423      Max.   :19.0757      Max.   :1.00000
## Phoneme.Inventory.Size Distance.to.Mainland Distance.to.Continent
## Min.      :2.398      Min.      :-0.6931      Min.      :-0.6931
## 1st Qu.:3.045      1st Qu.: -0.6931      1st Qu.: -0.6931
## Median :3.258      Median : -0.6931      Median : -0.6931
## Mean  :3.276      Mean  : 0.4460      Mean  : 0.5639
## 3rd Qu.:3.466      3rd Qu.: -0.6931      3rd Qu.: -0.6931
## Max.   :4.890      Max.   :15.3793      Max.   :15.5608
```

Adjusting NEE24 data with matrices

```
common_ids <- Reduce(intersect, list(
  nee24$ISO,
  rownames(phylomatrix),
  rownames(spmatrix)
))
nee24 <- nee24[nee24$ISO %in% common_ids, ]
phylomatrix <- phylomatrix[common_ids, common_ids]
spmatrix <- spmatrix[common_ids, common_ids]
```

Preview Data

```
paste("Size of dataset adjusted:", dim(nee24)[1])

## [1] "Size of dataset adjusted: 1716"

paste("#Island_Endemic: ", sum(nee24$Island.Endemic))

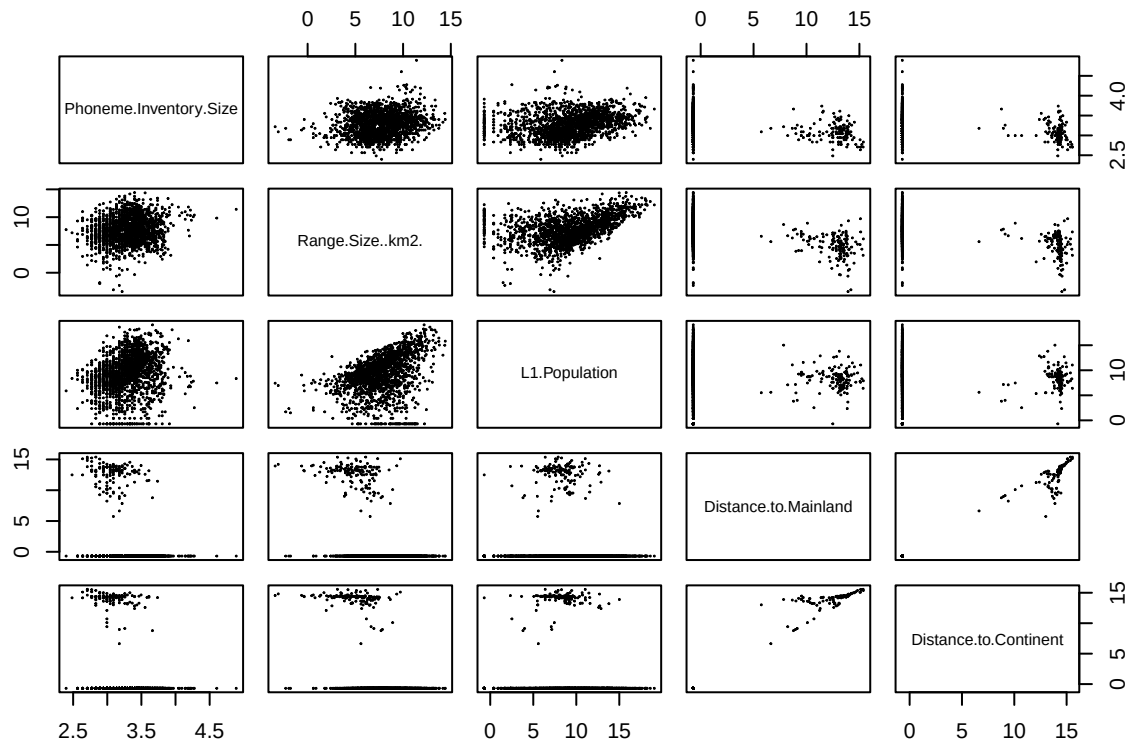
## [1] "#Island_Endemic: 153"

paste("#Island_Endemic to Total Ratio: ", sum(nee24$Island.Endemic) / dim(nee24)[1])

## [1] "#Island_Endemic to Total Ratio: 0.0891608391608392"
```

Variable Correlations

```
pairs(nee24[,
  c("Phoneme.Inventory.Size", "Range.Size..km2.",
    "L1.Population", "Distance.to.Mainland", "Distance.to.Continent")],
  panel = function(x, y, ...) {
    points(x, y, cex = 0.1, ...) # cex = 0.5 makes points half the default size
  })
```



```
cor(nee24[,
      c("Phoneme.Inventory.Size", "Range.Size..km2.",
        "L1.Population", "Distance.to.Mainland", "Distance.to.Continent")])
```

```
##              Phoneme.Inventory.Size Range.Size..km2. L1.Population
## Phoneme.Inventory.Size      1.0000000      0.1775584  0.22791579
## Range.Size..km2.            0.1775584      1.0000000  0.38202343
## L1.Population               0.2279158      0.3820234  1.00000000
## Distance.to.Mainland        -0.2057552     -0.3721935 -0.07647052
## Distance.to.Continent       -0.2067185     -0.3698787 -0.07452058
##              Distance.to.Mainland Distance.to.Continent
## Phoneme.Inventory.Size      -0.2057552      -0.20671848
## Range.Size..km2.            -0.37219347     -0.36987871
## L1.Population               -0.07647052     -0.07452058
## Distance.to.Mainland         1.00000000      0.99489824
## Distance.to.Continent        0.99489824      1.00000000
```

```
cor.test(nee24$Phoneme.Inventory.Size, nee24$Island.Endemic)
```

```
##
## Pearson's product-moment correlation
##
## data: nee24$Phoneme.Inventory.Size and nee24$Island.Endemic
## t = -8.807, df = 1714, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## -0.2529007 -0.1623492
## sample estimates:
## cor
## -0.2080707
```

Preview Matrices

```
print(dim(phylomatrix))
```

```
## [1] 1716 1716
```

```
print(dim(spmatrix))
```

```
## [1] 1716 1716
```

```
print(phylomatrix[1:10,1:10])
```

```
##      aak aaq aar aau abk      abm      abn abq      abr abt
## aak    1  0  0  0 0.0 0.0000 0.0000 0.0 0.0000 0
## aaq    0  1  0  0 0.0 0.0000 0.0000 0.0 0.0000 0
## aar    0  0  1  0 0.0 0.0000 0.0000 0.0 0.0000 0
## aau    0  0  0  1 0.0 0.0000 0.0000 0.0 0.0000 0
## abk    0  0  0  0 1.0 0.0000 0.0000 0.5 0.0000 0
## abm    0  0  0  0 0.0 1.0000 0.0782 0.0 0.0391 0
## abn    0  0  0  0 0.0 0.0782 1.0000 0.0 0.0391 0
## abq    0  0  0  0 0.5 0.0000 0.0000 1.0 0.0000 0
## abr    0  0  0  0 0.0 0.0391 0.0391 0.0 1.0000 0
## abt    0  0  0  0 0.0 0.0000 0.0000 0.0 0.0000 1
```

```
print(spmatrix[1:10,1:10])
```

```
##      aak      aaq      aar      aau      abk      abm      abn
## aak    0.0000 14533.967 11515.484  567.5621 11678.9524 15278.9681 15479.7010
## aaq 14533.9669      0.000 10112.225 14447.7100  7954.7468  8455.8165  8396.3112
## aar 11515.4836 10112.225      0.000 10943.3656  2803.7842  3448.0427  3685.6258
## aau  567.5621 14447.710 10943.366      0.0000 11055.6119 14747.4890 14960.8317
## abk 11678.9524  7954.747  2803.784 11055.6119      0.0000  5054.0987  5312.5124
## abm 15278.9681  8455.817  3448.043 14747.4890  5054.0987      0.0000  253.6514
## abn 15479.7010  8396.311  3685.626 14960.8317  5312.5124  253.6514      0.0000
## abq 11627.7133  7856.003  3110.033 11006.2051  119.4451  5273.4333  5527.2348
## abr 16341.3640  7392.015  4484.987 15798.9606  5600.8093  1052.1906  861.3983
## abt  453.9969 14334.058 11140.752  139.1289 11191.9302 14951.4017 15166.5587
##      abq      abr      abt
## aak 11627.7133 16341.3640  453.9969
## aaq  7856.0027  7392.0149 14334.0581
## aar  3110.0333  4484.9867 11140.7516
## aau 11006.2051 15798.9606  139.1289
## abk  119.4451  5600.8093 11191.9302
## abm  5273.4333  1052.1906 14951.4017
## abn  5527.2348  861.3983 15166.5587
## abq    0.0000  5771.1676 11138.9009
## abr  5771.1676    0.0000 16000.6922
## abt 11138.9009 16000.6922    0.0000
```

Analysis - Generalised Least Squares (GLS) (2)

GLS Setup (2)

```
best_p <- function (p,formula,data,spmatrix,phylomatrix) {
  spmatrix <- spmatrix/max(spmatrix)
```

```

    spmatrix <- exp(-(spmatrix/p[2])^2)
    mat <- as.matrix((p[3]*(1-p[1])*spmatrix+(1-p[1])*(1-p[3])*phyломatrix+p[1]*diag(dim(phyломatrix)[1],dim(phyломatrix)[2]))
    res <- try(gls(model=formula,data=data,correlation=corSymm(mat[lower.tri(mat)]),fixed=T),method="ML")
    if (inherits(res,"try-error")) {
      out <- -10000
    } else {
      out <- res$logLik
      if (res$logLik>0) {out <- -10000}
    }
    -out
  }

ml_fit <- function (p,formula,data,spmatrix,phyломatrix) {
  spmatrix <- spmatrix/max(spmatrix)
  spmatrix <- exp(-(spmatrix/p[2])^2)
  mat <- as.matrix((p[3]*(1-p[1])*spmatrix+(1-p[1])*(1-p[3])*phyломatrix+p[1]*diag(dim(phyломatrix)[1],dim(phyломatrix)[2]))
  res <- try(gls(model=formula,data=data,correlation=corSymm(mat[lower.tri(mat)]),fixed=T),method="ML")
  res
}

```

Model predictor combinations

```

predictors <- c("Island.Endemic", "Range.Size..km2.",
               "L1.Population", "Distance.to.Mainland", "Distance.to.Continent")
response <- "Phoneme.Inventory.Size"
n_pred <- length(predictors)
pred_combs <- sapply(1:n_pred, function(x) combn(predictors, x))

models <- c(Phoneme.Inventory.Size~1)
for (i in 1:n_pred) {
  #print(pred_combs[[i]])
  for (j in 1:dim(pred_combs[[i]])[2]) {
    model <- (paste(response, paste(pred_combs[[i]][,j], collapse="+"), sep="~"))
    model <- as.formula(model)
    models <- c(models,model)
  }
}
unlist(models)

```

```

## [[1]]
## Phoneme.Inventory.Size ~ 1
##
## [[2]]
## Phoneme.Inventory.Size ~ Island.Endemic
##
## [[3]]
## Phoneme.Inventory.Size ~ Range.Size..km2.
##
## [[4]]
## Phoneme.Inventory.Size ~ L1.Population
##
## [[5]]
## Phoneme.Inventory.Size ~ Distance.to.Mainland

```

```

##
## [[6]]
## Phoneme.Inventory.Size ~ Distance.to.Continent
##
## [[7]]
## Phoneme.Inventory.Size ~ Island.Endemic + Range.Size..km2.
##
## [[8]]
## Phoneme.Inventory.Size ~ Island.Endemic + L1.Population
##
## [[9]]
## Phoneme.Inventory.Size ~ Island.Endemic + Distance.to.Mainland
##
## [[10]]
## Phoneme.Inventory.Size ~ Island.Endemic + Distance.to.Continent
##
## [[11]]
## Phoneme.Inventory.Size ~ Range.Size..km2. + L1.Population
##
## [[12]]
## Phoneme.Inventory.Size ~ Range.Size..km2. + Distance.to.Mainland
##
## [[13]]
## Phoneme.Inventory.Size ~ Range.Size..km2. + Distance.to.Continent
##
## [[14]]
## Phoneme.Inventory.Size ~ L1.Population + Distance.to.Mainland
##
## [[15]]
## Phoneme.Inventory.Size ~ L1.Population + Distance.to.Continent
##
## [[16]]
## Phoneme.Inventory.Size ~ Distance.to.Mainland + Distance.to.Continent
##
## [[17]]
## Phoneme.Inventory.Size ~ Island.Endemic + Range.Size..km2. +
##     L1.Population
##
## [[18]]
## Phoneme.Inventory.Size ~ Island.Endemic + Range.Size..km2. +
##     Distance.to.Mainland
##
## [[19]]
## Phoneme.Inventory.Size ~ Island.Endemic + Range.Size..km2. +
##     Distance.to.Continent
##
## [[20]]
## Phoneme.Inventory.Size ~ Island.Endemic + L1.Population + Distance.to.Mainland
##
## [[21]]
## Phoneme.Inventory.Size ~ Island.Endemic + L1.Population + Distance.to.Continent
##
## [[22]]
## Phoneme.Inventory.Size ~ Island.Endemic + Distance.to.Mainland +

```

```

##      Distance.to.Continent
##
## [[23]]
## Phoneme.Inventory.Size ~ Range.Size..km2. + L1.Population + Distance.to.Mainland
##
## [[24]]
## Phoneme.Inventory.Size ~ Range.Size..km2. + L1.Population + Distance.to.Continent
##
## [[25]]
## Phoneme.Inventory.Size ~ Range.Size..km2. + Distance.to.Mainland +
##      Distance.to.Continent
##
## [[26]]
## Phoneme.Inventory.Size ~ L1.Population + Distance.to.Mainland +
##      Distance.to.Continent
##
## [[27]]
## Phoneme.Inventory.Size ~ Island.Endemic + Range.Size..km2. +
##      L1.Population + Distance.to.Mainland
##
## [[28]]
## Phoneme.Inventory.Size ~ Island.Endemic + Range.Size..km2. +
##      L1.Population + Distance.to.Continent
##
## [[29]]
## Phoneme.Inventory.Size ~ Island.Endemic + Range.Size..km2. +
##      Distance.to.Mainland + Distance.to.Continent
##
## [[30]]
## Phoneme.Inventory.Size ~ Island.Endemic + L1.Population + Distance.to.Mainland +
##      Distance.to.Continent
##
## [[31]]
## Phoneme.Inventory.Size ~ Range.Size..km2. + L1.Population + Distance.to.Mainland +
##      Distance.to.Continent
##
## [[32]]
## Phoneme.Inventory.Size ~ Island.Endemic + Range.Size..km2. +
##      L1.Population + Distance.to.Mainland + Distance.to.Continent

```

Fitting Models

Best p values for all models

```

# Get best p for all models with intercept only
p.res <- sbplx(c(0.5, 0.5, 0.5),
              best_p,
              formula=Phoneme.Inventory.Size~1,
              data=nee24,
              spmatrix=spmatrix, phylomatrix=phylomatrix,
              lower=c(0,0,0), upper=c(1,1,1),
              nl.info = TRUE)

```


Maximum likelihood fits for all models

```
lm.res_list <- c()

ml_fits <- mclapply(
  X = models,
  FUN = function(f) ml_fit(p=p.res$par, formula=f, data=nee24, spmatrix=spmatrix, phylomatrix=phylomatrix),
  mc.cores = 6
)
```

Model Comparison

References

1. Bromham, L., Yaxley, K.J. & Cardillo, M. Islands are engines of language diversity. *Nat Ecol Evol* **8**, 1991–2002 (2024). <https://doi.org/10.1038/s41559-024-02488-4>
2. Hua, X., Greenhill, S.J., Cardillo, M. *et al.* The ecological drivers of variation in global language diversity. *Nat Commun* **10**, 2047 (2019). <https://doi.org/10.1038/s41467-019-09842-2>